

Chapter 5:

Artificial Intelligence & Machine Learning

5.1 Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a major branch of computer science focused on creating intelligent systems and machines capable of thinking, reasoning, learning, solving problems, planning, understanding natural language, and acting autonomously like human beings.

Key Definition

Primary Goal of AI: To simulate human intelligence and solve complex, multi-faceted problems across various domains.

Applications of Artificial Intelligence

- a) Gaming:** AI models applied in chess, strategic games, and interactive gaming environments.
- b) Natural Language Processing (NLP):** Enables systems to parse, comprehend, and generate human languages.
- c) Automated Grading:** Automated assessment tools for educational tasks and exam evaluation.
- d) Personalized Learning:** Tailoring educational content according to individual student learning pace.
- e) Expert Systems:** Knowledge-based systems designed to emulate decision-making of human experts.
- f) Vision Systems:** Image processing, recognition, and scene interpretation.
- g) Handwriting Recognition:** Translating handwritten notes/text into digitized, actionable data.
- h) Intelligent Robots:** Autonomous robotic systems capable of navigating and performing tasks.

5.2 Introduction to Machine Learning

Machine Learning (ML) is a subfield of Artificial Intelligence that allows computers to learn directly from data and past experiences to improve their performance over time without being explicitly programmed for every single task.

Applications of Machine Learning

- a) Email and Spam Detection:** Filtering malicious or unwanted emails automatically.
- b) Speech Recognition:** Converting acoustic signals and spoken language into textual input.
- c) Image Recognition:** Identifying objects, facial features, and patterns within images.
- d) Self-Driving Cars:** Autonomous navigation using sensor data and ML models.
- e) Fraud Detection:** Detecting anomalous transactions in financial and banking systems.

- f) Social Media Feeds:** Curating customized content and recommendations on social platforms.
- g) Dark Web Monitoring:** Automated threat intelligence and security surveillance.
- h) Chatbots & Virtual Assistants:** Conversational AI agents for automated customer support.

5.3 Comparison: AI vs. Machine Learning

Artificial Intelligence (AI)	Machine Learning (ML)
Branch of computer science aimed at simulating human intelligence.	A subset of AI focused on learning from data to make predictions.
Simulate human intelligence and solve complex problems.	Enable computers to learn from patterns and past data automatically.
Uses data, rules, algorithms, and logical/heuristic methods.	Relies heavily on statistical data analysis and iterative training.
Can function on hardcoded rules without massive datasets.	Requires large volumes of training data to operate effectively.
Overarching field and ultimate design destination.	The engine and primary method used to achieve modern AI.

5.4 Machine Learning Paradigms/ Learning Techniques

Supervised Learning Techniques

In supervised learning, the machine is provided with training data along with guidance/labels. The algorithm is trained on input-output pairs with clear target answers or labels, enabling it to learn mapping functions for prediction and classification.

Unsupervised Learning Techniques

In unsupervised learning, the machine is not provided with labelled data. Instead, the algorithm must analyze raw inputs to find hidden patterns, clusters, or structural groupings by itself without human supervision.

Comparison: Supervised vs. Unsupervised Learning

Supervised Learning	Unsupervised Learning
The computer learns from labelled data.	The computer learns from unlabelled data.
It knows the correct output while training	It tries to find patterns or groups in the data.
It is like learning with teachers who give the right answers.	It is like learning without a teacher by exploring the materials.
Example: Email is classified as spam or not spam	Example: Grouping customers based on shopping pattern

5.5. AI in Robotics, Simulation of Simple Robotic Tasks

Artificial Intelligence (AI) in robotics is the study of designing, building, and operating robots. For example, they can be used for bomb disposal, exploring the ocean, fighting fires, space exploration, and working in nuclear or chemical environments.

Follow the given steps to create a simple robotic task.

Step 1: Visit the website <https://scratch.mit.edu/>

Step 2: Delete the default sprite. Choose a robot sprite with an appropriate backdrop.

Step 3: Drag the following code and click on the green flag.

Step 4: You will see the following result:



5.6 Generative AI & Tool Integration

Generative AI (Gen-AI) is an advanced computer technology capable of creating original content across various modalities, including text, images, music, audio, and video. It learns complex structures from massive datasets and uses this knowledge to produce new, synthetic output.

Applications of AI Integrated Tools

Application of integrated AI tools in google docs:

- i. Smart compose with auto correction
- ii. Grammar and Spelling
- iii. Voice typing
- v. Language translation
- vi. Auto summary

Application of integrated AI tools in email:

- i. Smart compose and quick reply
- ii. Schedule email
- iii. Spam filtering
- iv. Grammar and writing suggestions
- v. Email summarization

Application of integrated AI tools in Office 365

- i. Word AI powered writing tools
 - ii. AI driven data analysis in excel
 - iii. AI enhanced presentation
 - iv. AI powered meetings and collaboration
- Popular Generative AI Tools

Tool Name	Developed By / Company	Key Leadership / CEO
Copilot	OpenAI & GitHub / Microsoft	Satya Nadella
ChatGPT	OpenAI	Sam Altman
Gemini	Google DeepMind / Google	Sundar Pichai

5.7 Ethics, Security, and Emerging Technologies

- **Ethics in AI:** The set of moral principles that define right and wrong behavior in technology creation and deployment.
- **Bias in AI:** Unfair decisions or differential treatment of individuals based on features like gender, race, or socio-economic background.
- **Privacy in AI:** Keeping personal information and sensitive data private, secure, and protected against unauthorized access.
- **Security in AI:** Protecting AI infrastructure, pipelines, and algorithms from malicious threats, hacking, or data tampering.

5.8 Internet of Things (IoT)

IoT is a network of physical objects (vehicles, smart home devices, TVs, sensors) that collect data via sensors and exchange information across the Internet with minimal to no human intervention.

IoT Applications:

- | | |
|---------------------|--------------------------|
| i. Smart education | ii. Smart home |
| iii. Smart city | iv. Smart transportation |
| v. Smart healthcare | vi. Smart agriculture |
| vii. Communication | |



5.9 Virtual & Extended Reality (VR/XR)

Virtual Reality (VR): A computer-generated environment that immerses users, making it feel real using devices like VR headsets.

Extended Reality (XR): An umbrella term encompassing all immersive technologies that merge the physical and digital worlds. This includes

- **VR (Virtual Reality):** A complete, simulated virtual environment.
- **AR (Augmented Reality):** Overlays digital images, objects, or text onto the real world using phones or AR glasses.

Applications of VR/XR:

- a. Education
- b. Training
- c. Health and medication
- d. Business
- e. Entertainment
- f. Military and defense
- g. Tourism and hospitality
- h. Aeronautic

5.10 Cloud Computing

Cloud computing is a technology paradigm that allows users to access shared computing resources—such as servers, data storage, databases, networking, and software applications—on-demand over the Internet.

Examples & Benefits

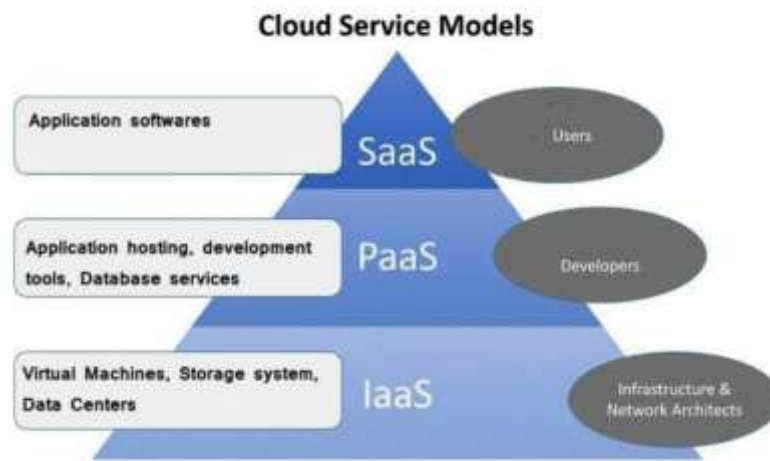
- i. Improves teamwork and productivity
- ii. Reduces hardware costs
- iii. Keeps data safe and always available
- iv. Provides access to advanced tools and services

Core Features of Cloud Computing

- 1. On-Demand Self-Service
- 2. Ubiquitous Access (Anytime, Anywhere)
- 3. Scalable Resources & Dynamic Elasticity
- 4. Shared Infrastructure & Multi-Tenancy
- 5. Robust Security Protocols
- 6. Easy Maintenance & Automated Updates
- 7. Low Capital & Operational Cost
- 8. Reliable Performance & High Availability

5.11 Cloud Service Models

Service Model	Full Name	Description & Capabilities	Popular Examples
SaaS	Software as a Service	Delivers software applications over the internet without needing local device installation.	Gmail, Google Docs, Microsoft Office 365
PaaS	Platform as a Service	Provides a cloud environment and framework to build, test, deploy, and manage software apps.	Google App Engine, Microsoft Azure App Services
IaaS	Infrastructure as a Service	Provides virtualized hardware resources (servers, storage, networks), eliminating physical management costs.	Google Cloud Storage, AWS EC2, OneDrive



5.12 E-Commerce, E-Government, and E-Education

Electronic Commerce (E-Commerce)

The process of buying, selling, or exchanging goods and services using electronic systems and the Internet.

- **Merits:** Online shopping convenience, time savings, 24/7 availability, access to global product catalogs, and doorstep delivery.
- **Limitations:** Product mismatch risk, online fraud, reduced social interaction, heavy technology dependency, and delivery delays.

Electronic Governance (E-Government)

The application of digital technologies and the Internet by public sector agencies to deliver public services and official information to citizens.

- **Merits:** Enhances public transparency, ensures administrative accountability, speeds up service delivery, provides equal accessibility, and reduces corruption.
- **Limitations:** Digital divide, technical system failures, and low digital literacy among certain demographics.

Electronic Education (E-Education)

The use of digital platforms, multimedia, and the Internet to facilitate teaching, learning, and academic interaction.

- **Merits:** Self-paced learning, access to diverse study materials, anytime access, cost affordability, and learning flexibility.
- **Limitations:** Reduced personal interaction, technical connectivity problems, and variable course content quality.